

AFIN8003 - Workshop 7

Banking and Financial Intermediation

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1 MCQ

1. A method of measuring loan concentration by tracking credit ratings of firms in particular classes for unusual declines is known as:
 - ☐ concentration risk
 - ☐ migration analysis
 - ☐ diversification analysis
 - ☐ minimum risk analysis
2. The measure used to reflect the historic experience of a pool of loans in terms of their credit-rating migration over time is a:
 - ☐ credit swap
 - ☐ migration matrix
 - ☐ concentration limit
 - ☐ minimum risk analysis
3. Management is unwilling to permit losses exceeding 5 per cent of an FI's capital, and it estimates that the amount lost per dollar of defaulted loans is 20 cents. What is the maximum amount of loans to a single sector as a per cent of capital?
 - ☐ 2.5%
 - ☐ 25%
 - ☐ 4%
 - ☐ 1%
4. The combination of assets that reduces the variance of portfolio returns to the lowest feasible level is the:
 - ☐ efficient frontier
 - ☐ least cost portfolio
 - ☐ minimum risk portfolio
 - ☐ maximum return portfolio
5. The risk of a loan (σ_i) is measured as
 - ☐ 1-LGD
 - ☐ 1-EDF
 - ☐ volatility of the default rate $\times$ LGD
 - ☐ $X_i - R_i$
6. An FI holds 500 residential mortgages of equal size and equal risk. As it keeps adding more mortgages of the same kind, portfolio risk:
 - ☐ falls to zero

- ☐ does not change
 - ☐ falls, but only to a floor set by the correlation between the loans
 - ☐ rises, because the book becomes more concentrated
7. A swap that involves swapping an obligation to pay interest, at a fixed or floating rate, for payments representing the total return on a loan or a bond of a specified amount is a(n):
- ☐ plain vanilla swap
 - ☐ off-market swap
 - ☐ fixed-floating swap
 - ☐ total return swap
8. Two loan portfolios both sit on the efficient frontier. Portfolio A has an expected return of 6 per cent and risk of 4 per cent. Portfolio B has an expected return of 9 per cent and risk of 8 per cent. With a risk-free rate of 2 per cent, which has the higher Sharpe ratio?
- ☐ A, because it earns more return per unit of risk
 - ☐ B, because its expected return is higher
 - ☐ They are equal, because both lie on the efficient frontier
 - ☐ It cannot be determined without the correlation between them

2 Short answer questions

2.1 Q1

How do loan portfolio risks differ from individual loan risks?

2.2 Q2

Why is it difficult for small banks, credit unions and building societies to measure credit risk using modern portfolio theory?

2.3 Q3

CountrySide Bank uses the Moody's Analytics RiskFrontier model to evaluate the risk–return characteristics of the loans in its portfolio. A specific \$10 million loan earns 2 per cent per year in fees, and the loan is priced at a 4 per cent spread over the cost of funds for the bank. For collateral considerations, the loss to the bank if the borrower defaults will be 20 per cent of the loan's face value. The expected probability of default is 3 per cent. What is the anticipated return on this loan? What is the risk of the loan?

2.4 Q4

BOQ's loan book at 31 August 2025 was 74 per cent residential mortgages, spread across tens of thousands of separate borrowers in several states.

- (a) An analyst argues that with so many individual borrowers the portfolio must be well diversified. Explain why that reasoning is incomplete.
- (b) Suppose each mortgage has a standard deviation of 4 per cent and the average pairwise correlation between them is 0.25. What is the lowest portfolio risk the bank could reach by adding more mortgages of the same kind?
- (c) Given your answer to (b), name two things the bank can do about the risk that remains.

2.5 Q5

How is selling a credit forward similar to buying a put option?